

CL1002 – Programming Fundamentals Lab



Lab # 01

Introduction to C Programming

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Programming:

Is the process of creating a set of instructions that tell a computer how to perform a task.

Introduction to C:

C is a general-purpose programming language developed in 1972 by Dennis Ritchie.

C has been used to write everything from operating systems (including Windows and many others) to complex programs like the Python interpreter, Git, Oracle database, and more.

C programming is considered as the base for other programming languages, most of the compilers, Kernels, etc. are written in C language, and most of the programming languages follow C syntax, for example, C++, Java, C#, etc. That is why it is known as mother language.

IDE/Tools:

1. Dev
2. Visual code
3. Visual studio

Installation procedure for visual studio code:

<https://www.javatpoint.com/how-to-run-a-c-program-in-visual-studio-code>

Structure of a C program:

We can formally assess the structure of a C program. By structure, it is meant that any program can be written in this structure only. Writing a C program in any other structure will hence lead to a Compilation Error.

The structure of a C program is as follows:

Structure of C Program

<i>Header</i>	<code>#include <stdio.h></code>
<i>main()</i>	<code>int main() {</code>
<i>Variable declaration</i>	<code>int a = 10;</code>
<i>Body</i>	<code>printf("%d ", a);</code>
<i>Return</i>	<code>return 0; }</code>



The components of the above structure are:

#include <stdio.h> includes the standard input output library functions. The printf() function is defined in stdio.h .

int main() The main() function is the entry point of every program in c language.

printf() The printf() function is used to print data on the console.

scanf() Takes the input from user

return 0 The return 0 statement, returns execution status to the OS. The 0 value is used for successful execution.

Format Specifiers:

Specifier	Output Type
%d or %i	A decimal integer or signed integer
%c	Signed character
%f	Signed float
%e	A floating-point number
%s	A string or sequence of character
%lf	double
%Lf	Long double
%o	Octal integer
%u	Short unsigned integer
%ld	Long decimal integer
%x	Hexadecimal integer
%p	Print memory address in the hexadecimal form

Escape Sequence:

Constant	Meaning
\a	Alert (bell, alarm)
\b	Backspace
\f	Form feed (new page)
\n	New-line
\r	Carriage return
\t	Horizontal tab
\v	Vertical tab
\'	Single quotation mark

\"	Double quotation mark
\?	Question mark
\\	Backslash

Variables in C:

- A named memory location where data is stored is called variable.
- A quantity whose value may change during execution of the program is called variable. It is represented by a symbol or name.
- Variable is name of reserved area allocated in memory. In other words, it is a name of memory location.
- No whitespace is allowed within the variable name.
- A variable name must not be any reserved word or keyword, e.g. int, float, etc.
- Variable name must be meaningful and should not be too long (convention).

Valid Variables:

- student_name
- _aSystemName
- A1
- stdNm
- _antMaN

Invalid Variables:

- \$sum
- 2names
- stu Nmbr
- int
- float

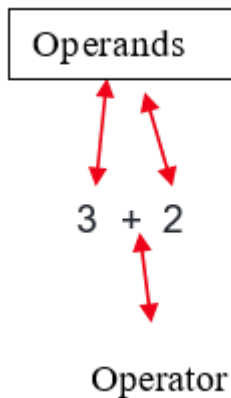
Data Types:

A data type, in programming, is a classification that specifies which type of value a variable has and what type of mathematical, relational or logical operations can be applied to it without causing an error.

- short int
- int
- long int
- float
- double
- long double
- char
- void

Operators

Operators are special symbols that carry out arithmetic or logical computation. The value that the operator operates on is called the operand.



Here, `+` is the operator that performs addition. `2` and `3` are the operands and `5` is the output of the operation.

```
int sum1 = 100 + 50;    // 150 (100 + 50)
int sum2 = sum1 + 250;  // 400 (150 + 250)
int sum3 = sum2 + sum2; // 800 (400 + 400)
```

Arithmetic Operators

Arithmetic operators are used to perform mathematical operations like addition, subtraction, multiplication etc.

Operator	Meaning	Example
+	Add two operands	$x + y$
-	Subtract right operand from the left	$x - y$
*	Multiply two operands	$x * y$
/	Divide left operand by the right one.	x / y
%	Modulus - remainder of the division of left operand by the right	$x \% y$ (remainder of x/y)

Assignment operators

Assignment operators are used to assign values to variables.

`int a = 5` is a simple assignment operator that assigns the value 5 on the right to the variable `a` on the left.

Operator	Example	Equivalent to
=	<code>x = 5</code>	<code>x = 5</code>
+=	<code>x += 5</code>	<code>x = x + 5</code>
-=	<code>x -= 5</code>	<code>x = x - 5</code>
*=	<code>x *= 5</code>	<code>x = x * 5</code>

/=	x /= 5	x = x / 5
%=	x %= 5	x = x % 5

Expression

An expression is a combination of operators, constants and variables.

Examples:

$x+y$

$x+y-1$

$x+1$

Precedence of Operators

There can be more than one operator in an expression.

To evaluate these types of expressions there is a rule of precedence. It guides the order in which these operations are carried out.

For example, multiplication has higher precedence than subtraction.

But we can change this order using parentheses () as it has higher precedence than multiplication.

Multiplication, division and modulus have the same priority but higher than addition and subtraction.

* / %

+ -

Example Program:

```
#include<stdio.h>
int main()
{
    int a;
    int b;
    printf("Enter the value of a: ");
    scanf("%d",&a);
    printf("\n");
    printf("Enter the value of b: ");
    scanf("%d",&b);
    int sum=a+b;
    int diff=a-b;
    int product=a*b;
    int division=a/b;
    int mod=a%b;
    printf("a = %d b = %d", a, b);
    printf("\nSum: %d", sum);
    printf("\nDiff: %d", diff);
    printf("\nProduct: %d", product);
    printf("\nDivision: %d", division);
    printf("\nModulus: %d", mod);
    return 0;
}
```

Lab Task:

1. Write a program to convert feet to inches
2. Write a program that takes 5 numbers from user and finds the average
3. Write a program to subtract the highest number from the average observed in task 2 (both values must be taken from task 2)
4. Write a program to find the distance between two points (The distance formula is derived from the Pythagorean theorem. To find the distance between two points (x_1, y_1) and (x_2, y_2) , all that you need to do is use the coordinates of these ordered pairs and apply the formula.)

Distance Formula:

Assume we have two points M and N, with coordinates (x_1, y_1) and (x_2, y_2) respectively. Their distance can be represented as MN and it can be calculated as given below formula,

The first point (M):- (x_1, y_1)

Second point (N):- (x_2, y_2)

Distance (MN):- $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

Example:-

M = (4, 8)

N = (12, 14)

Then distance between M and N is, $MN = \sqrt{((12-4)^2 + (14-8)^2)} = \sqrt{(64 + 36)} = \sqrt{(100)} =$

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- 5. Write a C program to compute $(a - b)^3$ where the value of a and b will be provided by the user.**

The formula of $(a - b)^3$ is given below.

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$